

Does a High Minimum Wage Insulate or Expose Workers to the Effects of Technological Change? Evidence from U.S. States

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Abstract

This paper examines whether state minimum wage increases change how technological change affects employment and wages in the United States. Using occupation-level automation exposure measures from Webb (2020) and exploiting cross-state variation in minimum wages from 1980 to 2010, we find that higher minimum wages magnify both the employment and wage losses associated with software exposure: software-exposed occupations fared worse in states where minimum wages increased the most. The pattern differs for industrial robots. We find no comparable employment effect, but wages decline more in robot-exposed occupations where minimum wages rose more, with a wage interaction about twice the magnitude of the software estimate. Taken together, the results indicate that minimum wages do not interact uniformly with automation: wage floors amplify broad employment and wage adjustment under software exposure but mainly wage adjustment under robots. Minimum wage policy alone therefore appears insufficient to shield workers from technological change, and its labor-market effects depend on the form of automation to which workers are exposed.

JEL Codes: J23, J31, J38, O33

Keywords: Minimum wage, automation, technological change, employment, wages, robots, artificial intelligence

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1 Introduction

Two major changes have shaped the U.S. labor market over the last four decades. First, automation has expanded rapidly: the number of industrial robots increased from a few thousand in 1980 to more than 250,000 by 2017, while software transformed many white-collar jobs (Acemoglu and Restrepo, 2020; Webb, 2020). Second, state minimum wage policies have increasingly diverged from the federal wage floor. In 1980, only two states had minimum wages above the federal level, but by 2017 that number had risen to 30 states, including the District of Columbia. In some states, such as Washington, the minimum wage reached \$11, and it reached \$11.50 in the District of Columbia. Several U.S. cities have also implemented higher local minimum wages, with Seattle and nearby Sea-Tac reaching \$15.¹ These parallel trends raise a central question: do higher minimum wages insulate workers from automation-related losses, or do they unintentionally increase the vulnerability of at-risk workers?

This question lies at the intersection of two main branches of labor economics. One literature documents the effects of automation on employment and wages, particularly for workers performing routine tasks (Autor, Levy, and Murnane, 2003; Acemoglu and Restrepo, 2020; Webb, 2020). Another large literature studies the effects of minimum wages, finding significant wage gains for low-wage workers and small, limited, or contested employment effects (Card and Krueger, 1994; Dube, Lester, and Reich, 2010; Cengiz et al., 2019; Manning, 2021). However, less is known about how minimum wage policies interact with technological change.

This paper studies whether minimum wage growth changes how automation exposure translates into employment and wage outcomes. The analysis is organized around two competing hypotheses. Under the “exposure” hypothesis, higher minimum wages increase labor-cost pressure in occupations where technology can substitute for workers. Minimum wage increases may therefore magnify the employment and wage losses associated with automa-

¹See <https://www.dol.gov/whd/minwage/america.htm> (viewed February 1, 2017).

tion exposure, leading to larger losses in more exposed occupations.² Under the “insulation” hypothesis, higher minimum wages protect workers in automation-exposed occupations by limiting wage erosion or supporting worker bargaining power. Minimum wage increases may therefore dampen the employment and wage losses associated with automation exposure, leading to smaller losses in more exposed occupations.³ The analysis is closely related to studies showing that minimum wage increases can shift employment away from automatable or routine-cognitive low-wage jobs (Lordan and Neumark, 2018; Aaronson and Phelan, 2019). This paper addresses two specific research questions. First, do higher state minimum wages amplify or dampen the employment and wage effects of automation exposure in the United States over 1980–2010, and are these effects stronger in states where minimum wages increased the most? Second, do these interaction effects differ across technology type, specifically between industrial robots, which are more closely linked to routine manual tasks in manufacturing, and software, which is more closely linked to routine cognitive tasks in clerical and administrative occupations?

Using the occupation-level exposure measures developed by Webb (2020), which quantify the extent to which industrial robots and software can substitute for human labor in different occupations, we combine state-level variation in minimum wage policies from 1980 to 2010 with U.S. Census and American Community Survey (ACS) data aggregated to the occupation-industry-state level. We employ long-difference regressions to analyze the combined effects of automation exposure and minimum wage growth on employment and wages over time.

Our empirical approach relies on variation along two dimensions: cross-state heterogeneity in minimum wage policy and occupation-level variation in automation exposure. The parameter of interest identifies whether occupations more exposed to robots or software exhibit different employment and wage responses in states where minimum wages rose the most. To account for potential confounders, we incorporate controls for offshorability, terciles of

²See Lordan and Neumark (2018) and Aaronson and Phelan (2019).

³See Card and Krueger (1994), Dube, Lester, and Reich (2010), Cengiz et al. (2019), and Manning (2021).

average years of education, initial wages, in addition to industry and state fixed effects. The specification further distinguishes between industrial robots and software because these technologies target different tasks and may therefore have different relationships with minimum wage policies.

Our results show three main findings. First, higher minimum wages magnify the employment and wage losses associated with software exposure. In the main specification for 1980–2010, the interaction between minimum wage increases and software exposure implies that states where minimum wages increased the most experienced larger employment and wage losses in software-exposed occupations (Tables 1 and 2). This pattern is consistent with task-based models of technological change, in which computer-based technologies substitute for routine, codifiable tasks (Autor, Levy, and Murnane, 2003), and with evidence that minimum wage increases can shift employment away from automatable or routine-cognitive low-wage jobs (Lordan and Neumark, 2018; Aaronson and Phelan, 2019).

Second, we find different results for industrial robots. The interaction between robot exposure and minimum wage increases is not associated with comparable employment losses in manufacturing occupations. For wages, however, the interaction is negative in the main specification, implying larger wage declines in robot-exposed occupations in states where minimum wages increased more (Tables 3 and 4). The robot wage interaction is also larger in magnitude than the corresponding software estimate: in the main 1980–2010 specification, it is about twice as large in absolute value. Therefore, the main findings reveal employment and wage losses for software-exposed occupations, but wage losses alone for robot-exposed occupations.

Third, we show that the labor-market effects of wage floors differ across technology types. A broad minimum wage literature finds that minimum wage increases raise wages for low-wage workers, while their aggregate employment effects are often small, limited, or contested (Card and Krueger, 1994; Cengiz et al., 2019; Manning, 2021). Dustmann et al. (2022) also show that minimum wages can reshape the allocation of workers across firms

without necessarily reducing employment. Our study is closely related to work showing that minimum wage increases can displace workers from automatable or routine-cognitive low-wage jobs (Lordan and Neumark, 2018; Aaronson and Phelan, 2019), while separately examining software and industrial robot exposure for both employment and wage outcomes.

This paper also speaks to the broader literature on technology and inequality. Acemoglu and Restrepo (2022) argue that automation accounts for 50–70% of the rise in U.S. wage inequality by displacing workers from routine tasks. Our findings imply that wage floors may interact with this process unevenly across technologies, reinforcing the importance of distinguishing between different forms of automation when evaluating labor-market policy.

The findings have policy implications, but they are specific rather than general. They do not imply that higher minimum wages have uniform labor-market effects in automatable jobs. Instead, the results show that minimum wage increases magnify the employment and wage losses associated with software exposure, while comparable employment declines are not detected for robot-exposed manufacturing occupations. This suggests that the interaction between minimum wage policy and automation depends on the type of technology to which workers are exposed.

The findings also suggest that minimum wage policies alone may be insufficient to protect workers from the disruptive effects of technological change. As automation changes labor demand, policies that help workers move across tasks, firms, and occupations may remain important. However, evidence on the success of workforce adjustment and retraining programs is limited (Hyman, 2022), highlighting that the design of effective policy for an era of technological transformation remains an important area for future work.

Although this paper does not estimate the labor-market effects of recent generative artificial intelligence directly, it speaks to an important question in the AI literature: how labor-market institutions shape the consequences of task-replacing technologies. This connection is natural because the automation exposure measures used in this paper come from Webb (2020), whose framework links patents to occupational task descriptions to measure

exposure to software, industrial robots, and artificial intelligence.

The relevance to AI is strongest through the task-based framework. In models of automation and AI, technology affects labor demand by replacing workers in some tasks, raising productivity in others, and potentially creating new tasks (Acemoglu and Restrepo, 2019; Acemoglu, 2024). The empirical results in this paper show that minimum wage increases do not interact uniformly with different technologies.

This distinction matters for thinking about AI. Recent AI systems, especially large language models, are closer to software-based automation than to industrial robots because they operate through information processing, writing, prediction, classification, coding, and other cognitive tasks. Recent studies find broad occupational exposure to these technologies (Eloundou et al., 2023; Felten, Raj, and Seamans, 2023), while firm-level evidence shows that AI exposure is associated with changes in hiring and skill requirements even when aggregate employment effects are still difficult to detect (Acemoglu et al., 2022).

The results should therefore be interpreted as a framework for studying AI, not as a forecast of AI’s labor-market effects. The evidence suggests that the consequences of wage floors may depend on the type of technology reshaping labor demand and on the tasks through which that technology operates. As AI diffuses across clerical, administrative, professional, and service occupations, future work should examine whether minimum wage policy amplifies, dampens, or redirects AI-related labor-market adjustment.

The rest of the paper is structured as follows. Section II describes the automation and minimum wage literatures. Section III explains the research methodology. Section IV presents the main empirical findings on employment and wages. Section V concludes.

2 Literature Review

2.1 Automation and Labor Markets

An expanding literature investigates how automation shapes employment and wage outcomes. Acemoglu and Restrepo (2020) offer a seminal framework that distinguishes automation technologies that substitute for labor and new task creation, which generates fresh labor demand. Using the geographic variation in industrial robot penetration from 1990–2007, they estimate that one additional robot per thousand workers decreases the employment-to-population ratio by 0.18–0.34 percentage points and wages by 0.25–0.50 percent. Their task-based approach underscores that automation’s net impact relies on the balance between job displacement and reinstatement forces.

Autor, Levy, and Murnane (2003) introduces the task-based approach to understanding technological change, showing that computerization displaces routine tasks while complementing non-routine cognitive tasks. This theory anticipates — with strong empirical support — that routine middle-skill occupations bear the brunt of displacement pressure. Acemoglu and Autor (2011) build on this foundation, demonstrating that skill-biased technological change and labor market polarization are interconnected trends driven by routine task automation.

Recent research has improved our knowledge of which technologies displace specific worker groups. Webb (2020) constructs occupation-level automation exposure for artificial intelligence, robotics, and software by matching patent text with the O*NET database of task descriptions. He documents substantial differences: AI disproportionately affects high-skill occupations, while robots primarily affect manufacturing and manual tasks. Crucially, Webb finds that occupations highly exposed to automation experienced a fall in both employment and wages, confirming automation exposure’s predictive validity.

Acemoglu and Restrepo (2022) provide an extensive analysis of automation’s influence on the US labor market from 1980–2016. They attribute that more than 50 percent of the rise in

wage inequality during this period operates chiefly through diminishing demand for middle-skill workers. Their findings further demonstrate that there are substantial differences across industries and time periods, with the overall effect depending on the relative magnitude of task displacement and generation of new tasks that restores labor demand.

The related strand of literature examined specific automation technologies more closely. Autor (2015) provides a historical and forward-looking perspective on workplace automation, documenting employment polarization and exploring the dual role of automation in substituting and complementing labor. To untangle the effects of trade and technology on manufacturing employment, Autor, Dorn, and Hanson (2015) use local labor market data, demonstrating that both contribute to job displacement via separate mechanisms. Dauth et al. (2021) study cross-country panel data and show that although industrial robots raise labor and total factor productivity, they reduce employment among low-skilled workers.

2.2 Minimum Wage Effects on Employment and Wages

The study of minimum wages and their effect on employment outcomes is among the most studied questions in labor economics. The standard competitive model predicts that binding minimum wages will lower employment (Stigler, 1946). Yet, the empirical findings have consistently questioned this prediction, in particular following Card and Krueger (1994)'s seminal study of the 1992 New Jersey minimum wage hike. Employing Pennsylvania as a control group, they find no detectable employment losses in the fast food industry, implying that monopsony power or other labor market frictions may matter considerably.

Neumark and Wascher (2008) offer a comprehensive assessment of the minimum wage literature and argue for a modest adverse impact on employment, in particular for teenagers and low-skilled workers. Nevertheless, they acknowledge considerable variation in findings across different studies and contexts. Subsequent work by Dube, Lester, and Reich (2010) highlights the importance of appropriate control groups, revealing that research using adjacent counties as controls yields smaller or insignificant employment effects in comparison to

those with wider geographic controls.

Cengiz et al. (2019) apply a bunching estimator to examine minimum wage changes from 1979–2016, demonstrating that minimum wages raise the earnings of low-wage workers with negligible employment effects. By focusing on the wage distribution rather than just aggregate employment, their approach provides a more nuanced view of minimum wage impacts. Similarly, Clemens and Wither (2021) argue that minimum wages improve compensation for employed workers but may substantially reduce employment prospects for the least-skilled workers.

The distributional impact of minimum wage policies has attracted significant scholarly attention. Lee (1999) demonstrates that minimum wages compress wage dispersion in the lower tail of the distribution, reducing wage inequality among low-wage workers. Recent empirical work examining minimum wages in specific settings has identified positive wage effects alongside minimal employment responses, while noting substantial heterogeneity depending on industry and region.

2.3 Technology and Minimum Wage Interaction

Although an extensive literature is available separately for automation and minimum wages, there is a dearth of studies that examined the combined effect of automation and minimum wages. Lordan and Neumark (2018) demonstrate that a 10 percent increase in minimum wages results in a fall in the share of automatable jobs held by low-skilled workers by 0.31 percentage points (elasticity = -0.10), and the effects are more prominent in manufacturing industries (elasticity = -0.18) as well as for older, female, and Black workers. Whilst, Aaronson and Phelan (2019) show a relative decline in employment as well as wages in cognitively routine occupations without affecting manual routine and non-routine employment with the increase in minimum wages. Theoretically, Eckardt (2022), using a task-based framework, demonstrates that the distributional consequences of minimum wages depend critically on the type of automation and which tasks are displaced, providing theoretical grounding for

the empirical heterogeneity documented across studies.

The current literature has two important limitations. Previous empirical studies did not use state minimum wage variation with technology-specific exposure measures separating industrial robots, which affect routine manual tasks, from software, which affects cognitive routine tasks. Moreover, no prior study has interacted Webb (2020)’s technology-specific occupation-level exposure measures with state minimum wage variation to examine the wage effects and differential impacts of distinct automation technologies on employment and earnings of the workers. Our paper addresses both gaps directly.

3 Data and Empirical Strategy

3.1 Data Sources

For our analysis, we use multiple data sources to construct occupation-industry-state cells covering 1980 through 2010. The state variation is essential for identification purposes: it allows us to use differences in minimum wage policies across states while holding fixed occupation-level technological exposure to automation.

3.1.1 Automation Exposure Measures

The automation exposure measures come from Webb (2020), who develops occupation-level indices for robots, software and AI exposure. Webb’s method uses the text of patent document filed with the U.S. Patent and Trademark Office and matches it to O*NET database of occupational task descriptions. The method extracts verb-noun pairs of patent abstracts and occupational task descriptions and then measures the overlap using text similarity algorithm. The resulting exposure indices capture each occupation’s technological potential for automation: higher exposure scores indicate occupations whose tasks more closely match the tasks described in automation-related patents.

Webb’s measures offer three advantages for our analysis. First, they improve upon ear-

lier routine-task measures (Autor, Levy, and Murnane, 2003) by directly linking specific technologies – software and industrial robots – to occupational tasks. Second, the exposure measures are predetermined relative to the outcomes studied here because they capture technological potential rather than firms’ actual adoption choices. This helps reduce concerns about reverse causality. Third, distinguishing among technology types allows us to examine whether software and industrial robots interact differently with minimum wage policy.

Our empirical analysis focuses on two automation types: software and industrial robots. Software exposure measures the feasibility of automating cognitive and clerical tasks, affecting occupations such as bookkeepers, data-entry workers, and administrative assistants. Robot exposure captures the potential for automating routine manual production tasks, particularly in manufacturing occupations.

3.1.2 Employment and Wage Data

We use individual-level microdata from the U.S. Census and the American Community Survey (ACS), provided by IPUMS USA (Ruggles et al., 2024), to construct employment and wage measures. Although we use several census years for robustness, the main analysis covers the 1980–2010 period. Following Webb (2020), the sample includes individuals aged 18–65 who were employed in the prior year, excluding self-employed workers, unpaid family workers, and military personnel. Occupations are classified using the `occ1990dd` scheme developed by Dorn (2009) and extended by Deming (2017), while industries follow the IPUMS `ind1990` classification.

The data construction mirrors Webb’s methodology. Labor-supply weights are constructed as full-time-equivalent (FTE) employment for each cell.⁴ Real weekly earnings, in 2016 dollars and deflated using the PCE price index, are computed for full-time, full-year (FTFY) workers, defined as those working more than 35 hours per week and at least 40

⁴Following Acemoglu and Autor (2011), IPUMS person weights (`PERWT`) are multiplied by each individual’s fraction of full-time work hours, so that the aggregated weights reflect FTE employment rather than raw headcounts.

weeks per year. To reduce the influence of outliers, wage observations are winsorized at the 98th percentile within each year.

The key innovation in our data construction is state-level disaggregation. Unlike Webb (2020), who analyzes occupation-industry-year cells at the national level, we disaggregate the data to occupation-industry-state-year cells. This generates cross-state variation in employment and wage outcomes that can be matched to differences in state minimum wage policies.

The final dataset contains substantial variation along both dimensions of the empirical design. Across occupations, software exposure is higher in clerical and administrative occupations, while robot exposure is concentrated in manufacturing production occupations. Across states, minimum wage growth varies considerably between 1980 and 2010, reflecting differences in whether states followed the federal floor or maintained higher state minimum wages.

3.1.3 Minimum Wage Data

The state and federal minimum wage data come from the U.S. Department of Labor. For each state, the effective minimum wage is the maximum of the applicable state and federal minimum wage. For the regression analysis, minimum wages are deflated to 2016 dollars using the PCE price index, consistent with the wage deflator used for earnings. Minimum wage changes exhibit substantial variation across states. Alaska and Connecticut were the only states with minimum wages above the federal minimum in 1980. By 2010, 15 states, including Washington, D.C., had minimum wages above the federal floor.

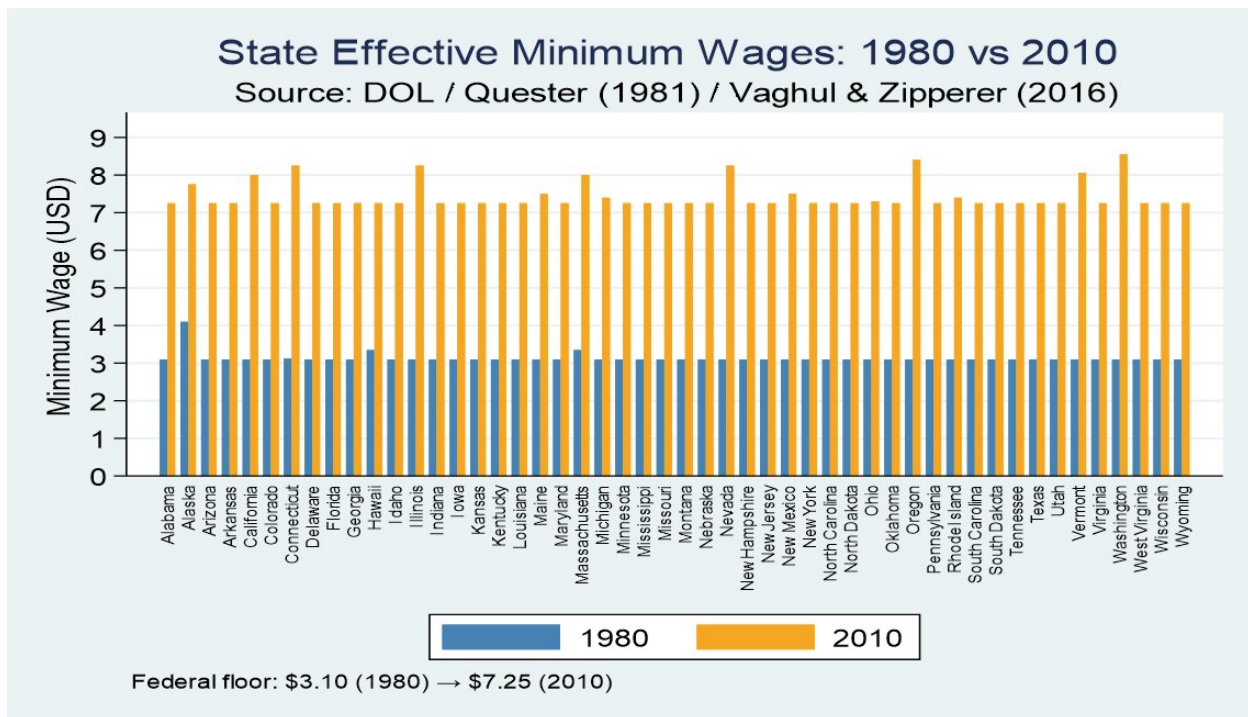


Figure 1: **State Effective Minimum Wages: 1980 versus 2010.** Each bar plots the state’s effective minimum wage—the maximum of the applicable state and federal floors—in nominal dollars. The federal floor rose from \$3.10 in 1980 to \$7.25 in 2010. Most states remained at or near the federal floor in 1980, while by 2010 a subset of states had enacted wage floors substantially above the federal minimum, generating the cross-state variation in minimum wage levels used for identification. *Source:* U.S. Department of Labor; Quester (1981); Vaghul and Zipperer (2016).

3.1.4 Control Variables

Occupation-industry controls are based on 1980 values to avoid conditioning on post-treatment outcomes. Education controls are labor-supply-weighted terciles of average years of schooling. We include 1980 mean log weekly wages for FTFY workers and its quadratic term to account for mean reversion and nonlinear relationships between baseline wages and subsequent changes. The offshorability measure comes from Firpo, Fortin, and Lemieux (2011) and is standardized by Autor and Dorn (2013). It is an occupation-level index capturing the potential for work to be performed remotely, based on whether an occupation requires face-to-face interaction or site-specific presence.

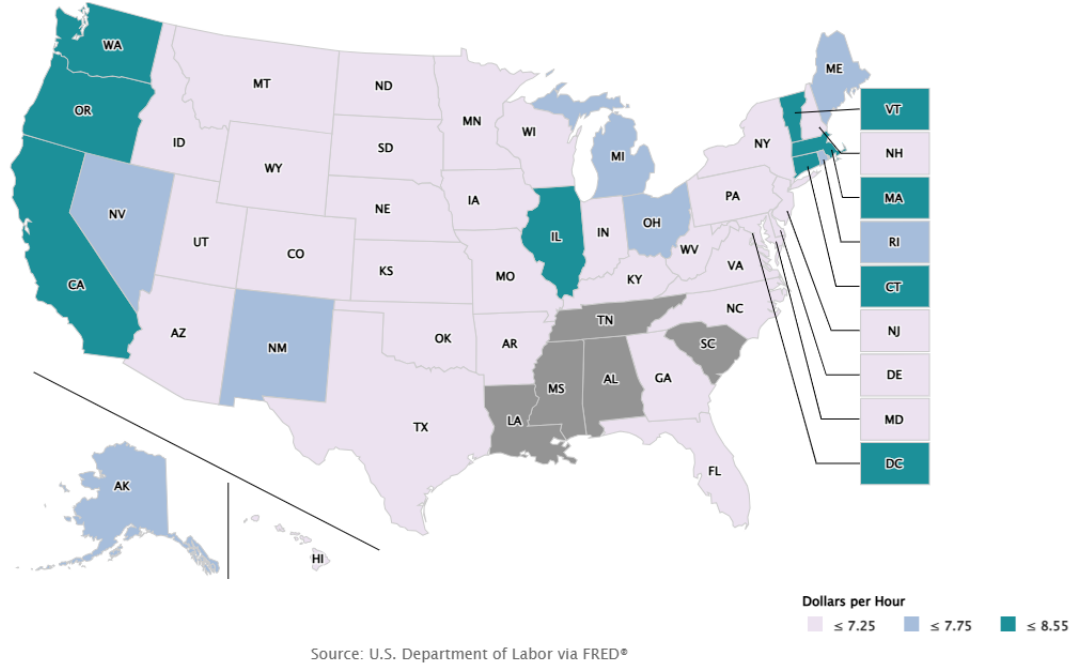


Figure 2: **Geographic distribution of state minimum wage rates, 2010 (dollars per hour).** The map displays the effective state minimum wage rate as of 2010 across all 50 states and the District of Columbia, grouped into three tiers: states at the federal floor (\$7.25, light purple), states modestly above the floor (\$7.26–\$7.75, light blue), and states substantially above the floor (\$7.76–\$8.55, teal). The geographic clustering is notable: high-wage states concentrate along the Pacific Coast (Washington, Oregon, California), in New England (Vermont, Massachusetts, Connecticut, Maine), and in the Mid-Atlantic corridor (DC, Maryland, New Jersey), while most Southern and interior states remained at the federal floor. *Source:* U.S. Department of Labor via FRED®.

3.1.5 Labor-Market Characteristics

In additional specifications, we also account for state labor-market institutions that may shape the relationship between minimum wages and automation exposure. These controls include right-to-work laws and unionization, measured at the state level. Right-to-work laws are coded using state adoption histories and capture whether a state limited union-security agreements during the sample period. Unionization measures come from Unionstats, which reports union membership and coverage rates using CPS data.

Because the relevance of unionization differs across technologies, we match the union-

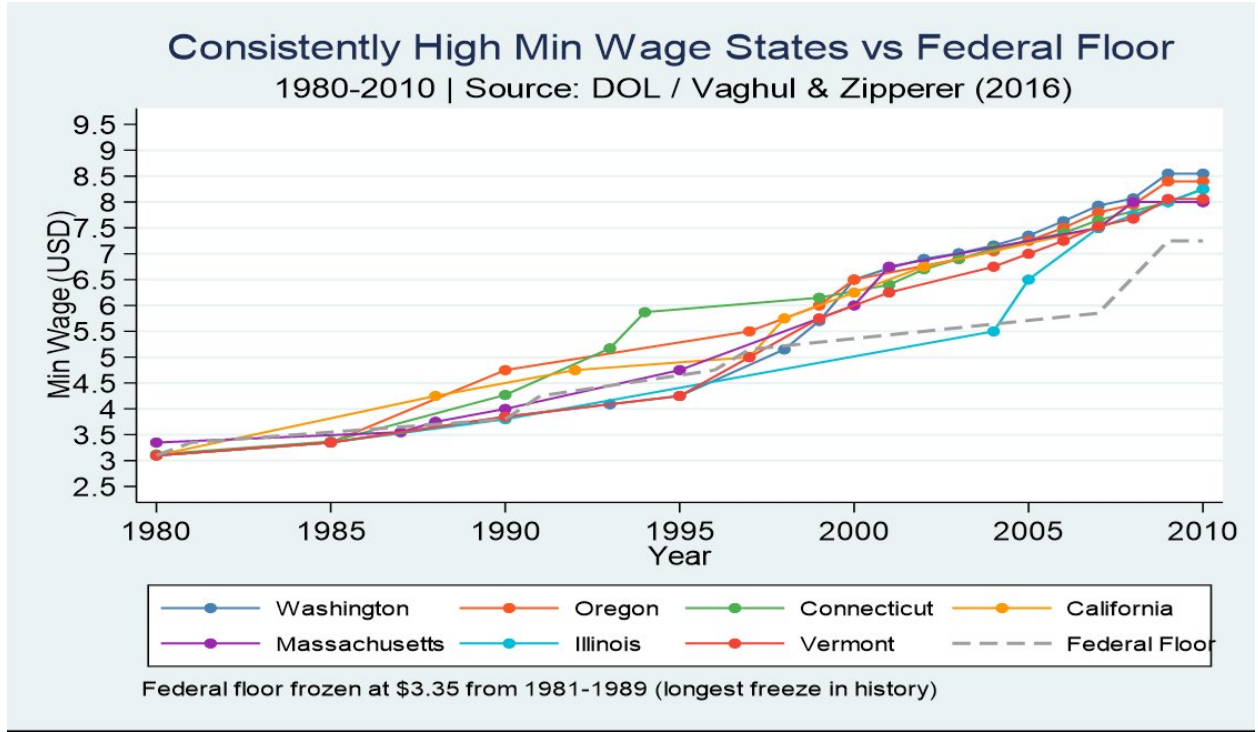


Figure 3: **Minimum wages in consistently high-wage states versus the federal floor, 1980–2010.** The figure traces the nominal minimum wage trajectory for states that persistently exceeded the federal floor over the sample period (Washington, Oregon, Connecticut, California, Massachusetts, Illinois, Vermont) alongside the federal floor (dashed). The federal floor was frozen at \$3.35 from 1981 through 1989—the longest such freeze in U.S. history—generating substantial and persistent real minimum wage dispersion across states. This variation, together with subsequent federal increases and further state-level activism, provides the identifying cross-state policy variation used in the empirical strategy. *Source:* U.S. Department of Labor; Vaghul and Zipperer (2016).

ization measure to the sector most closely related to each automation type. For the robot analysis, we use private manufacturing union coverage, since industrial robot exposure is concentrated in manufacturing. For the software analysis, we use private-sector union coverage, since software exposure is spread more broadly across clerical and administrative occupations outside manufacturing. We use baseline unionization and changes in unionization over the sample period to examine whether the main interaction between minimum wages and automation is sensitive to these state institutional characteristics.

Unionstats state-sector data begin in 1983, so we use 1983 as the baseline year for the

unionization measures. This is the closest available year to 1980 in a consistent state-sector series. The use of 1983 rather than 1980 reflects data availability rather than a change in the empirical design.

3.2 Empirical Strategy

We examine how state minimum wages influence the labor-market consequences of automation exposure using a long-difference approach over 1980–2010. We regress changes in employment and wages for occupation-industry-state cells on the interaction between state minimum wage increases and occupation-level automation exposure, along with state and industry fixed effects. This interaction term is the core of the identification strategy: it asks whether the same degree of automation exposure is associated with different employment and wage changes in states where minimum wages increased more.

The specification separately interacts minimum wage increases with exposure to industrial robots and exposure to software. This distinction is important because the two technologies substitute for different types of tasks: routine manual tasks concentrated in manufacturing occupations for robots, and routine cognitive tasks concentrated in clerical and administrative occupations for software. By examining both employment and wages, the analysis distinguishes whether minimum wages affect the number of workers in automation-exposed occupations, the wages of workers who remain, or both.

$$\Delta Y_{ois} = \alpha_i + \delta_s + \beta_1 \text{Exposure}_o + \beta_3 (\text{Exposure}_o \times \Delta \text{MinWage}_s) + \gamma' Z_{oi} + \epsilon_{ois} \quad (1)$$

The variable ΔY_{ois} measures the change in employment share or log wages for occupation o , industry i , and state s over 1980–2010. The change in employment share is measured using the Davis-Haltiwanger-Schuh arc percentage change (Davis, Haltiwanger, and Schuh, 1996), which accommodates entry and exit of occupation-industry-state cells. Exposure_o is the

standardized automation exposure measure from Webb (2020). $\Delta\text{MinWage}_s$ is defined as 100 times the log change in the real state minimum wage between 1980 and 2010. The vector Z_{oi} comprises education terciles measured in 1980, log wages and its quadratic term, and offshorability. α_i denotes dummy for industry fixed effects and δ_s denotes state fixed effects dummy.

The key coefficient of interest is β_3 . A negative coefficient implies that minimum wage increases are associated with larger employment or wage losses in automation-exposed occupations, consistent with the exposure hypothesis. A positive coefficient implies that minimum wage increases are associated with smaller employment or wage losses in automation-exposed occupations, consistent with the insulation hypothesis.

In additional specifications, we augment equation (1) with interactions between automation exposure and state labor-market institutions. These include baseline right-to-work status, changes in right-to-work status, baseline union coverage, and changes in union coverage. These specifications are used to assess whether the main interaction between minimum wage increases and automation exposure is sensitive to state institutional characteristics. We treat these results as additional evidence rather than the main specification.

4 Results

This section presents the key findings on the interaction between minimum wages and automation exposure. We first discuss software exposure, followed by industrial robot exposure. For each technology, we report results for employment and wages. In interpreting the tables, we discuss both the coefficient on automation exposure and the interaction between automation exposure and minimum wage increases. The exposure coefficient captures the baseline relationship between automation exposure and changes in labor-market outcomes, while the interaction coefficient shows whether this relationship differs in states where minimum wages increased more.

4.1 Software Exposure: Employment and Wage Effects

Table 1 presents the employment results for software exposure over 1980–2010. The coefficient on software exposure is negative across specifications, indicating that occupations more exposed to software automation experienced larger employment declines over the period. The estimate is large and precisely estimated in the simpler specifications, although it becomes smaller and less precise after the full set of controls is added.

The interaction between software exposure and minimum wage increases is negative and statistically significant across all specifications that include the interaction term. In column (3), after adding industry and state fixed effects, the interaction coefficient is -0.0061 with a standard error of 0.0018 . The estimate remains nearly unchanged after adding offshorability and education controls. In the full specification in column (6), the interaction coefficient is -0.0057 with a standard error of 0.0018 . This indicates that minimum wage increases magnify the employment losses associated with software exposure: software-exposed occupations experienced larger employment declines in states where minimum wages increased more. The employment results therefore support the exposure hypothesis for software automation.

This employment result is consistent with evidence that minimum wage increases can

shift employment away from automatable or routine-cognitive jobs. Lordan and Neumark (2018) find that minimum wage increases reduce the share of employment held by low-skilled workers in automatable occupations, while Aaronson and Phelan (2019) show that minimum wage hikes lead to relative employment declines in cognitively routine low-wage occupations.

Table 2 presents the corresponding wage results. The coefficient on software exposure is negative and statistically significant in all specifications, indicating that software-exposed occupations experienced larger wage declines over 1980–2010. In the full specification, the exposure coefficient is -0.2035 with a standard error of 0.0281 .

The interaction between software exposure and minimum wage increases is also negative and precisely estimated in every wage specification. The coefficient is -0.0056 in column (1), remains close to -0.0060 after adding fixed effects and additional controls, and equals -0.0050 in the full specification. This indicates that minimum wage increases also magnify the wage losses associated with software exposure. Together with the employment results, the software estimates support the exposure hypothesis: rather than insulating workers in software-exposed occupations, higher minimum wages are associated with larger employment and wage losses.

These findings are consistent with the broader automation literature showing that technological change has contributed to wage declines and rising inequality among workers performing routine tasks. Acemoglu and Restrepo (2022) argue that automation accounts for a substantial share of the rise in U.S. wage inequality since 1980. The results here point to a related channel: wage losses associated with software exposure are larger in states where minimum wages increased more.

Several features of these findings merit discussion. First, the interaction coefficient is negative even though the main effect of minimum wage increases on wages is positive in the specification without fixed effects. This suggests that software-exposed occupations face different wage dynamics than less-exposed occupations. Second, the education controls are positive and statistically significant in the wage regressions, indicating that human capital

is associated with stronger wage growth over the period. Third, the increase in adjusted R^2 after adding baseline wage controls underscores the importance of accounting for initial compensation levels.

Table 1: Employment Change vs. Software Exposure $\times$ Minimum Wage, 1980–2010

	(1)	(2)	(3)	(4)	(5)	(6)
Exposure	−0.3824*** (0.0266)	−0.2178*** (0.0800)	−0.2462*** (0.0772)	−0.2551*** (0.0793)	−0.1671* (0.0889)	−0.1505 (0.0909)
Min wage		−0.5210*** (0.1498)				
Exposure $\times$ Min wage			−0.0061*** (0.0018)	−0.0061*** (0.0018)	−0.0057*** (0.0019)	−0.0057*** (0.0018)
Offshorability				−2.2572 (1.9214)	−1.3981 (1.6463)	−0.7524 (1.7372)
Medium education					−5.6829* (3.2074)	−6.4771* (3.2922)
High education					21.5192*** (6.3737)	17.2626*** (5.9447)
Wage						0.0008 (0.0076)
Wage squared						0.0000** (0.0000)
Adjusted R^2	0.017	0.325	0.356	0.356	0.370	0.372
Industry FEs		Yes	Yes	Yes	Yes	Yes
State FEs			Yes	Yes	Yes	Yes
Observations	70,012	70,012	70,012	70,012	70,012	70,012

Notes: Each observation is an occupation-industry-state cell. Dependent variable is $100 \times$ DHS change of a cell's share of overall employment between 1980 and 2010, winsorized at the top and bottom 1%. Exposure is an occupation-level software exposure from Webb (2020). Minimum wage is the change in state minimum wage. Education variables are terciles of average years of education for occupation-industry cells in 1980. Wages are cells' mean weekly wage for full-time, full-year workers in 1980. Offshorability is an occupation-level measure from Autor and Dorn (2013). Observations are weighted by cell's labor supply, averaged between 1980 and 2010. Standard errors are clustered two-way by industry and state. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 2: Wage Change vs. Software Exposure $\times$ Minimum Wage, 1980–2010

	(1)	(2)	(3)	(4)	(5)
Exposure	−0.2071*** (0.0427)	−0.1659*** (0.0216)	−0.1655*** (0.0205)	−0.1326*** (0.0204)	−0.2035*** (0.0281)
Min wage	0.3134** (0.1293)				
Exposure $\times$ Min wage	−0.0056*** (0.0008)	−0.0059*** (0.0006)	−0.0059*** (0.0006)	−0.0060*** (0.0006)	−0.0050*** (0.0010)
Offshorability			0.0923 (0.7920)	0.3713 (0.7097)	−4.2440*** (0.6958)
Medium education				6.5394*** (0.9357)	12.1899*** (1.5371)
High education				6.7555*** (1.8508)	31.0250*** (1.7618)
Wage					−0.0788*** (0.0064)
Wage squared					0.0000*** (0.0000)
Adjusted R^2	0.018	0.177	0.177	0.181	0.334
Industry FEs		Yes	Yes	Yes	Yes
State FEs		Yes	Yes	Yes	Yes
Observations	63,143	63,143	63,143	63,143	63,143

Notes: Each observation is an occupation-industry-state cell. Dependent variable is $100 \times$ change in log wage between 1980 and 2010, winsorized at the top and bottom 1%. Exposure is an occupation-level software exposure from Webb (2020). Minimum wage is the change in state minimum wage. Education variables are terciles of average years of education for occupation-industry cells in 1980. Wages are cells' mean weekly wage for full-time, full-year workers in 1980. Offshorability is an occupation-level measure from Autor and Dorn (2013). Standard errors are clustered two-way by industry and state. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4.2 Robot Exposure: Employment and Wage Effects

Table 3 presents the employment results for robot exposure in manufacturing. The coefficient on robot exposure is negative and statistically significant across all specifications, indicating that manufacturing occupations more exposed to industrial robots experienced larger employment declines over 1980–2010. In the full specification in column (6), the exposure coefficient is -0.2360 with a standard error of 0.0544 .

In contrast, the interaction between robot exposure and minimum wage increases is not statistically significant in any employment specification. The coefficient is -0.0041 after adding industry and state fixed effects, and falls to -0.0016 in the full specification. These estimates indicate that, although robot exposure itself is associated with employment losses in manufacturing, minimum wage increases do not appear to magnify those employment

losses. The robot employment results therefore do not provide evidence for the exposure hypothesis along the employment margin.

This pattern is consistent with the distinction between the direct effect of robot exposure and the interaction between robot exposure and minimum wage policy. The negative exposure coefficients align with evidence that industrial robots reduce employment in exposed labor markets (Acemoglu and Restrepo, 2020). However, the insignificant interaction terms suggest that minimum wage increases do not significantly change the relationship between robot exposure and employment in manufacturing.

Table 4 presents the corresponding wage results. The coefficient on robot exposure is negative and statistically significant in all specifications, indicating that robot-exposed manufacturing occupations also experienced larger wage declines over 1980–2010. In the full specification, the exposure coefficient is -0.5285 with a standard error of 0.0340 .

The interaction between robot exposure and minimum wage increases is negative and statistically significant in the wage regressions. The coefficient is -0.0143 in column (1), remains close to -0.012 after adding fixed effects and additional controls, and equals -0.0103 in the full specification. This indicates that minimum wage increases amplify the wage losses associated with robot exposure: robot-exposed manufacturing occupations experienced larger wage declines in states where minimum wages increased more.

The magnitude of the robot wage interaction is also larger than the corresponding software estimate. In the full specification, the robot interaction coefficient is about twice as large as the software interaction coefficient, suggesting that minimum wage increases are associated with stronger wage losses in robot-exposed manufacturing occupations than in software-exposed occupations.

Taken together, the robot results differ from the software results. For robots, exposure is strongly associated with both employment and wage losses, but minimum wage increases magnify only the wage losses, not the employment losses. This suggests that the interaction between minimum wages and automation depends on the technology and the margin of

adjustment.

Table 3: Employment Change vs. Robot Exposure $\times$ Minimum Wage, 1980–2010

	(1)	(2)	(3)	(4)	(5)	(6)
Exposure	−0.5746*** (0.0341)	−0.5210*** (0.0718)	−0.5504*** (0.0788)	−0.5433*** (0.0794)	−0.2214*** (0.0451)	−0.2360*** (0.0544)
Min wage		−0.6152 (0.4008)				
Exposure $\times$ Min wage			−0.0041 (0.0046)	−0.0041 (0.0046)	−0.0015 (0.0047)	−0.0016 (0.0046)
Offshorability				0.9438 (1.0757)	−0.3473 (1.0941)	−0.7548 (1.2685)
Medium education					−2.0803 (2.4536)	−1.8841 (2.4293)
High education					37.5727*** (5.9436)	37.6194*** (6.0276)
Wage						−0.0198* (0.0113)
Wage squared						0.0000*** (0.0000)
Adjusted R^2	0.039	0.209	0.257	0.257	0.283	0.286
Industry FEs		Yes	Yes	Yes	Yes	Yes
State FEs			Yes	Yes	Yes	Yes
Observations	21,686	21,686	21,686	21,686	21,686	21,686

Notes: Each observation is an occupation-industry-state cell. Dependent variable is $100 \times$ DHS change of a cell's share of overall employment between 1980 and 2010, winsorized at the top and bottom 1%. Exposure is occupation-level robot exposure from Webb (2020). Minimum wage is the change in the minimum wage. Education variables are terciles of average years of education for occupation-industry cells in 1980. Wages are cells' mean weekly wage for full-time, full-year workers in 1980. Offshorability is an occupation-level measure from Autor and Dorn (2013). Observations are weighted by cell's labor supply, averaged between 1980 and 2010. Sample is restricted to industries within the manufacturing sector. Standard errors are clustered two-way by industry and state. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Wage Change vs. Robot Exposure $\times$ Minimum Wage, 1980–2010

	(1)	(2)	(3)	(4)	(5)
Exposure	−0.3841*** (0.0389)	−0.3202*** (0.0306)	−0.3026*** (0.0324)	−0.2504*** (0.0302)	−0.5285*** (0.0340)
Min wage	0.8139* (0.4161)				
Exposure $\times$ Min wage	−0.0143*** (0.0047)	−0.0125*** (0.0040)	−0.0124*** (0.0041)	−0.0123*** (0.0041)	−0.0103** (0.0043)
Offshorability			2.3423*** (0.6896)	2.3433*** (0.6887)	−5.9991*** (0.5527)
Medium education				4.0380*** (0.7603)	7.6123*** (0.9857)
High education				5.3081** (2.2618)	34.0236*** (2.4612)
Wage					−0.0911*** (0.0063)
Wage squared					0.0000*** (0.0000)
Adjusted R^2	0.044	0.128	0.130	0.132	0.318
Industry FEs		Yes	Yes	Yes	Yes
State FEs		Yes	Yes	Yes	Yes
Observations	19,736	19,736	19,736	19,736	19,736

Notes: Each observation is an occupation-industry-state cell. Dependent variable is $100 \times$ change in log wage between 1980 and 2010, winsorized at the top and bottom 1%. Exposure is an occupation-level robot exposure from Webb (2020). Minimum wage is the change in state minimum wage. Education variables are terciles of average years of education for occupation-industry cells in 1980. Wages are cells' mean weekly wage for full-time, full-year workers in 1980. Offshorability is an occupation-level measure from Autor and Dorn (2013). Sample is restricted to industries within the manufacturing sector. Standard errors are clustered two-way by industry and state. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4.3 State Labor-Market Institutions

We next examine whether the main interaction between minimum wages and automation exposure is sensitive to state labor-market institutions. These specifications add interactions between automation exposure and right-to-work status, changes in right-to-work status, baseline union coverage, and changes in union coverage. Because the regressions include state fixed effects, these state-level institutional variables cannot be included separately. They enter instead through their interactions with automation exposure.

For the software specifications, we use private-sector union coverage. For the robot specifications, we use private manufacturing union coverage because the robot analysis is restricted to manufacturing. Union coverage is measured from Unionstats state-sector data. Since the state-sector series begins in 1983, we use 1983 as the baseline unionization measure

and 2010 as the endpoint. Thus, these controls should be interpreted as sector-matched institutional controls measured near the start and end of the 1980–2010 period.

Table 5 compares the coefficient on the minimum-wage-by-automation interaction with and without these institutional controls. The software results are stable. For employment, the interaction coefficient is -0.0057 without institutional controls and -0.0088 after adding them. For wages, the corresponding coefficients are -0.0050 and -0.0045 . Both remain statistically significant. These estimates indicate that the software results are not driven by differences across states in right-to-work status or private-sector union coverage.

The robot results are different. The employment interaction remains statistically insignificant after adding institutional controls. The wage interaction is negative and statistically significant in the main specification, but it becomes smaller and statistically insignificant once right-to-work status and private manufacturing union coverage are included. This suggests that the wage result for robot exposure is more sensitive to state manufacturing labor-market institutions than the corresponding software result.

Overall, the institutional-control specifications reinforce the main conclusion for software exposure: minimum wage increases are associated with larger employment and wage losses in software-exposed occupations. For robots, the evidence remains weaker. Robot exposure itself is negatively related to manufacturing employment and wages, but the additional interaction with minimum wages is not robust once sector-matched state institutions are included.

Table 5: State Labor-Market Institution Controls

	Software exposure		Robot exposure	
	Employment (1)	Wages (2)	Employment (3)	Wages (4)
Exposure	−0.2098** (0.1001)	−0.1992*** (0.0610)	−0.1263 (0.1331)	−0.7528*** (0.0562)
Exposure × minimum wage increase	−0.0088*** (0.0023)	−0.0045*** (0.0012)	−0.0059 (0.0061)	−0.0056 (0.0041)
Exposure × RTW 1980	−0.0071 (0.0437)	−0.0003 (0.0228)	−0.0951 (0.0756)	0.1266** (0.0516)
Exposure × change in RTW	0.1358*** (0.0480)	0.0798** (0.0350)	−0.0190 (0.1970)	0.0212 (0.0501)
Exposure × union coverage, 1983	0.0148*** (0.0020)	−0.0026 (0.0036)	0.0053 (0.0064)	0.0102*** (0.0033)
Exposure × change in union coverage	0.0218*** (0.0047)	−0.0042 (0.0046)	0.0139* (0.0078)	0.0058 (0.0068)
Industry fixed effects	Yes	Yes	Yes	Yes
State fixed effects	Yes	Yes	Yes	Yes
Baseline controls	Yes	Yes	Yes	Yes
Adjusted R^2	0.372	0.334	0.285	0.319
Observations	70,012	63,143	21,686	19,736

Notes: Standard errors are in parentheses. The software columns use private-sector union coverage. The robot columns use private manufacturing union coverage. Union coverage is measured in 1983 and 2010; the change in union coverage is 2010 minus 1983. Baseline controls include offshorability, 1980 education terciles, 1980 wages, and 1980 wages squared. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4.4 Robustness: Wage Band Restriction

We next restrict the sample to occupation-industry-state cells whose baseline weekly wages fall in the part of the wage distribution most relevant for minimum wage policy. Specifically, we keep cells with 1980 weekly wages between 274.09 and 1,252.98. The lower bound corresponds to the 1980 federal minimum wage converted to 2016 dollars using the PCE deflator and applied to a 35-hour workweek, while the upper bound is four times the corresponding full-time weekly minimum wage. This restriction leaves 58,717 observations.

Table 6 shows that the main results remain similar in this restricted sample. For software exposure, the interaction between software exposure and minimum wage increases re-

mains negative for both employment and wages. In the full specification, the coefficients are -0.0062 for employment and -0.0050 for wages. For industrial robots, the employment interaction remains small and statistically insignificant, while the wage interaction remains negative and statistically significant. This pattern is consistent with the main estimates: minimum wage increases are associated with both employment and wage adjustment in software-exposed occupations, while the corresponding robot interaction appears mainly through wages.

Table 6: Wage Band Decomposition, 1980–2010

	Software exposure		Robot exposure	
	Employment (1)	Wages (2)	Employment (3)	Wages (4)
Exposure	-0.1383 (0.1006)	-0.1821^{***} (0.0302)	-0.2743^{***} (0.0606)	-0.5196^{***} (0.0277)
Exposure $\times$ minimum wage increase	-0.0062^{***} (0.0018)	-0.0050^{***} (0.0009)	-0.0024 (0.0046)	-0.0076^{**} (0.0032)
Industry fixed effects	Yes	Yes	Yes	Yes
State fixed effects	Yes	Yes	Yes	Yes
Baseline controls	Yes	Yes	Yes	Yes
Adjusted R^2	0.378	0.321	0.274	0.308
Observations	58,717	52,892	17,793	16,123

Notes: Standard errors are in parentheses. The sample is restricted to occupation-industry-state cells with 1980 weekly wages between \$274.09 and \$1,252.98. The lower bound is the 1980 federal minimum wage converted to 2016 dollars using the PCE deflator and applied to a 35-hour workweek. The upper bound is four times the corresponding full-time weekly minimum wage. Robot specifications are restricted to manufacturing. Baseline controls include offshorability, 1980 education terciles, 1980 wages, and 1980 wages squared. Minimum wage increase is measured as the change in log real state minimum wage multiplied by 100. Standard errors are clustered two-way by state and industry. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

4.5 Common Support Across States

We next examine whether the restricted-sample results are driven by occupation-industry cells that are observed in only a small number of states. These cells provide limited common support because they do not allow meaningful comparisons across states with different minimum wage increases. We therefore remove occupation-by-industry cells that are present in at most five states.

This restriction removes 1,773 observations and leaves 56,944 observations, or 96.98 percent of the minimum-wage-relevant sample. The results are nearly unchanged. The software interaction remains negative and statistically significant for both employment and wages, with full-specification coefficients of -0.0060 and -0.0050 , respectively. For industrial robots, the estimates are also almost unchanged after imposing the common-support restriction. The employment interaction remains small and statistically insignificant, while the wage interaction remains negative and close to the restricted-sample estimate, with a coefficient of -0.0074 .

Overall, the common-support restriction does not materially change the results. The estimates remain very close to those obtained in the minimum-wage-relevant sample, indicating that the findings are not driven by occupation-industry cells that are present in only a small number of states.

Table 7: Common Support Across States

	Wage band restricted sample		Drop cells in ≤ 3 states		Drop cells in ≤ 5 states	
	Coef.	Obs.	Coef.	Obs.	Coef.	Obs.
Software exposure $\times$ minimum wage increase, employment	-0.0062^{***} (0.0018)	58,717	-0.0061^{***} (0.0018)	57,717	-0.0060^{***} (0.0017)	56,944
Software exposure $\times$ minimum wage increase, wages	-0.0050^{***} (0.0009)	52,892	-0.0050^{***} (0.0009)	52,136	-0.0050^{***} (0.0009)	51,526
Robot exposure $\times$ minimum wage increase, employment	-0.0024 (0.0046)	17,793	-0.0025 (0.0046)	17,642	-0.0024 (0.0046)	17,560
Robot exposure $\times$ minimum wage increase, wages	-0.0076^{**} (0.0032)	16,123	-0.0075^{**} (0.0031)	16,007	-0.0074^{**} (0.0031)	15,940

Notes: Standard errors are in parentheses. The table reports the coefficient on automation exposure interacted with minimum wage increase under alternative common-support restrictions. The first two columns use the wage band restricted sample from Table 6. Columns 3–4 drop occupation-by-major-industry cells observed in three or fewer states. Columns 5–6 drop occupation-by-major-industry cells observed in five or fewer states. All specifications use the full set of baseline controls, industry fixed effects, and state fixed effects. Robot specifications are restricted to manufacturing. Standard errors are clustered two-way by state and industry. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

4.6 Heterogeneity by Baseline Wage

We next examine whether the interaction effects differ across the baseline wage distribution within the minimum-wage-relevant sample. We divide cells into three groups using survey-weighted terciles of 1980 weekly wages: low-wage cells (274 – -567), middle-wage cells

(567 – –857), and high-wage cells (857 – –1,253). The same cutoffs are applied to both the full sample used for software exposure and the manufacturing sample used for robot exposure.

This exercise helps assess whether the interaction effects are concentrated in parts of the wage distribution where minimum wage policy is likely to bind more strongly. It also allows the software and robot results to differ in ways consistent with task-based models of automation: software is more closely related to routine cognitive tasks, while industrial robots are concentrated in routine manual tasks in manufacturing (Autor, Levy, and Murnane, 2003; Webb, 2020; Acemoglu and Restrepo, 2020).

The results are consistent with this distinction. For industrial robots, the wage interaction is largest in the low-wage group and becomes smaller at higher baseline wages. This pattern is consistent with a minimum-wage-bite interpretation: among manufacturing cells exposed to robots, wage adjustment is strongest where baseline wages are closer to the minimum-wage-relevant part of the distribution. For software exposure, the interaction is not concentrated only among the lowest-wage cells. The wage interaction is small in the low-wage group but negative in the middle and high wage groups, suggesting that software-related adjustment is not limited to the bottom of the wage distribution.

These estimates should be interpreted as descriptive heterogeneity rather than as a separate identification strategy. Because the groups are defined using baseline wages and one of the outcomes is wage growth, differences across wage bands may partly reflect mean reversion. The results are therefore best read as evidence on where the main estimates are concentrated, not as separate estimates that identify independent effects within each wage group.

Appendix Tables 9–15 provide an occupation-level diagnostic decomposition of the interaction coefficients. The decomposition shows that the negative wage interactions are not driven only by high-wage occupations. In the low- and middle-wage part of the 1980 wage distribution, the main downward contributors include machine operators, janitors, laborers,

packers, secretaries, cashiers, food preparation workers, and related production and service occupations. This pattern supports the interpretation that the estimated interactions are relevant to the minimum-wage-affected part of the labor market. These tables are descriptive diagnostics rather than separate identification exercises.

Table 8: Heterogeneity by Baseline Wage With the restricted Sample

Outcome	Wage band	Obs.	Exposure	Exposure $\times$ minimum wage increase	Adjusted R^2	Difference vs. low	p -value
<i>Panel A. Software exposure and employment</i>							
	Low	20,437	−0.1347 (0.1594)	−0.0052 (0.0100)	0.463	ref.	
	Middle	20,783	−0.1387 (0.1163)	−0.0043 (0.0054)	0.370	+0.0003 (0.0135)	0.983
	High	17,495	−0.1567 (0.1058)	−0.0100** (0.0045)	0.376	−0.0009 (0.0105)	0.932
<i>Panel B. Software exposure and wages</i>							
	Low	17,707	−0.0618 (0.0505)	−0.0008 (0.0022)	0.337	ref.	
	Middle	18,925	−0.2672*** (0.0320)	−0.0065** (0.0026)	0.297	−0.0061 (0.0045)	0.181
	High	16,255	−0.2175*** (0.0262)	−0.0043*** (0.0014)	0.377	−0.0035 (0.0021)	0.103
<i>Panel C. Robot exposure and employment</i>							
	Low	4,579	−0.4359*** (0.1266)	−0.0044 (0.0177)	0.376	ref.	
	Middle	6,837	−0.2835*** (0.0726)	−0.0212*** (0.0065)	0.269	−0.0093 (0.0180)	0.605
	High	6,376	−0.1323* (0.0714)	0.0035 (0.0079)	0.288	+0.0129 (0.0149)	0.391
<i>Panel D. Robot exposure and wages</i>							
	Low	3,995	−0.6786*** (0.0161)	−0.0211*** (0.0077)	0.307	ref.	
	Middle	6,212	−0.5152*** (0.0375)	−0.0086** (0.0038)	0.243	+0.0128 (0.0092)	0.167
	High	5,913	−0.4688*** (0.0379)	−0.0018 (0.0031)	0.315	+0.0179* (0.0090)	0.051

Notes: Standard errors are in parentheses. Wage bands are survey-weighted terciles of 1980 weekly wages within the wage band sample: low wage (\$274–\$567), middle wage (\$567–\$857), and high wage (\$857–\$1,253). The same cutoffs are applied to the full sample used for software exposure and to the manufacturing sample used for robot exposure. Each row reports the full specification within a wage band. Difference vs. low reports the coefficient on the triple interaction between exposure, minimum wage increase, and the wage-band indicator, with the low-wage group as the omitted category. These estimates are descriptive heterogeneity checks because wage groups are defined using baseline wages. Standard errors are clustered two-way by state and industry. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4.7 Interpretation and Mechanisms

The results point to a specific form of interaction between minimum wages and automation exposure. The coefficient on automation exposure captures whether more exposed occupations experienced different employment or wage changes over time, while the interaction

term asks whether this relationship differs in states where minimum wages increased more. The estimates therefore speak to whether minimum wage increases amplify or dampen the labor-market effects associated with automation exposure.

The main results are most consistent with the exposure hypothesis for software automation. In states where minimum wages increased more, occupations more exposed to software experienced larger declines in both employment and wages. This pattern suggests that minimum wage increases may strengthen adjustment pressures in occupations where routine cognitive tasks can be reorganized or replaced by software. The result does not require software-exposed workers to be minimum-wage workers themselves. A more plausible interpretation is that many software-exposed occupations perform tasks that are complementary to lower-wage labor. When minimum wages increase, firms may reorganize production, reduce reliance on routine labor, or adopt software-based systems that change demand for related clerical, administrative, sales, and support occupations. Under this mechanism, the demand shift can affect workers above the minimum-wage margin.

The robot results suggest a different adjustment margin. The interaction between minimum wage increases and robot exposure does not generate a comparable employment decline in manufacturing occupations. However, the wage interaction is negative in the main specification, indicating weaker wage growth in robot-exposed occupations in states where minimum wages increased more. In the main specification, the wage interaction for robot exposure is about twice as large in absolute value as the corresponding wage interaction for software exposure. Thus, for industrial robots, the additional adjustment associated with minimum wage increases appears mainly on the wage margin rather than through measured employment losses.

This difference between software and robots is economically meaningful. Software can be adopted across a broad set of industries and can reorganize routine cognitive tasks in many occupations. Robots, by contrast, operate through a more capital-intensive manufacturing margin, where adoption is tied to plant-level organization, production scale, and tasks suited

to physical automation.

The wage-band and occupation-level diagnostics provide additional context for this interpretation. The negative wage interactions are accounted for in part by occupations in the low- and middle-wage parts of the wage distribution, including production, service, sales, clerical, and administrative occupations. At the same time, higher-wage occupations also contribute in some specifications. Therefore, the diagnostics should not be read as evidence that the mechanism operates only among workers directly paid near the minimum wage. Rather, they show that the estimated interaction reflects a broader set of occupations connected to routine tasks and to the organization of low-wage work.

Overall, the evidence suggests that minimum wage increases do not interact uniformly with automation. The software results support the exposure hypothesis: higher minimum wages are associated with larger employment and wage losses in software-exposed occupations. The robot results show a narrower pattern, with no comparable employment effect but a negative wage interaction in the main specification. These findings imply that the labor-market consequences of minimum wage policy depend not only on the level of the wage floor, but also on the type of technology reshaping labor demand.

5 Conclusion

This study examines whether higher minimum wages insulate workers from the effects of technological change or expose them to greater labor-market adjustment. Using occupation-level automation exposure measures from Webb (2020) and cross-state variation in minimum wages over 1980–2010, we compare automation-exposed occupations across states where minimum wages increased the most. We find evidence more consistent with the exposure hypothesis. Minimum wage increases are associated with larger employment and wage losses in occupations exposed to software automation. For industrial robots, we find no comparable employment effect, although wages decline is larger in magnitude in robot-exposed

occupations in comparison to software in the main specification.

The results show that minimum wages do not interact uniformly with automation. For software exposure, the interaction appears on both the employment and wage margins. This is consistent with task-based models in which computer-based technologies substitute for routine cognitive tasks (Autor, Levy, and Murnane, 2003; Webb, 2020). For robot exposure, the main adjustment appears on the wage margin rather than through measured employment losses. This distinction is consistent with evidence that robots and software affect different parts of the task distribution and may therefore have different labor-market consequences (Acemoglu and Restrepo, 2020; Webb, 2020).

The interpretation is not that all automation-exposed workers are directly paid the minimum wage. Rather, minimum wage increases may affect occupations connected to low-wage labor through production reorganization, changes in task demand, or technology adoption. This mechanism is related to evidence that minimum wage increases can accelerate substitution away from automatable low-wage work (Lordan and Neumark, 2018; Aaronson and Phelan, 2019). Our results extend this line of evidence by showing that the interaction differs across software and industrial robot exposure.

The findings contribute to the literature on minimum wages, automation, and inequality by showing that labor-market institutions may shape how technological change is transmitted to workers. A large literature finds that minimum wages raise earnings for many low-wage workers, often with limited aggregate employment effects (Card and Krueger, 1994; Cengiz et al., 2019; Manning, 2021). Our results suggest that these average effects can mask important differences across occupations exposed to different forms of automation. They also speak to work showing that automation has contributed substantially to changes in the wage structure and inequality (Acemoglu and Restrepo, 2022).

Given the structure of the data, several limitations warrant attention. The analysis is conducted at the occupation-industry-state level and does not follow individual workers or firms over time. As a result, the estimates cannot distinguish whether the observed wage

changes reflect real-wage declines for workers who remain in the same occupation, changes in the types of workers employed within occupation-industry-state cells, or transitions of workers across occupations and industries. Longitudinal worker-level data would make it possible to separate within-worker wage changes from sorting and compositional change.

Administrative data with information on hours, earnings, employers, and job transitions would also help identify the margins along which firms and workers adjust. Such data could show whether the estimated effects operate through hours, movement across firms, changes in occupation, or changes in the types of workers employed in automation-exposed jobs. Finally, applying this framework to newer measures of artificial intelligence exposure would be useful as AI technologies spread across white-collar, professional, and service occupations.

Overall, the evidence indicates that minimum wage increases may amplify labor-market adjustment in some automation-exposed occupations rather than uniformly protect workers from technological change. This does not imply that minimum wages are ineffective as labor-market policy. Instead, it suggests that wage floors operate within a changing technological environment, and their effects may depend on the tasks and technologies that shape labor demand.

A Appendix

A.1 Occupation-Level Drivers of the Interaction Effects

The tables in this subsection provide a diagnostic decomposition of the automation exposure by minimum-wage interaction coefficients. The exercise uses the Frisch-Waugh-Lovell representation of the coefficient after residualizing the outcome and interaction term with respect to the same controls and fixed effects used in the main specification. It is not a separate identification strategy. Instead, it shows which occupations contribute most to the estimated interaction coefficients.

The decomposition is useful for checking whether the negative interactions are driven by occupations plausibly relevant to minimum-wage policy. The main diagnostic therefore separates low- and middle-wage occupations from high-wage occupations using weighted 1980 weekly wages.

Audit. Table 9 verifies that the occupation-level contributions sum back to the estimated interaction coefficient in each specification.

Exposure rankings. Table 10 reports the occupations with the highest automation exposure. This table is useful for describing which occupations are most exposed, but exposure alone does not determine which occupations drive the interaction coefficient.

Low- and middle-wage downward contributors. Table 11 is the central diagnostic table. It restricts attention to occupations whose weighted mean 1980 weekly wage is in the low or middle part of the minimum-wage distribution. For software, the downward contributors include low- and middle-wage occupations such as farm workers, cashiers, food preparation workers, janitors, laborers, packers, and secretaries. For robots, the wage result is driven mainly by middle-wage manufacturing and manual occupations such as machine operators, packers, janitors, laborers, and welders.

Table 9: FWL Diagnostic Audit

Specification	Interaction coefficient	Sum of occupation contributions	Difference	Occupations
Robot x employment	−0.002442	−0.002442	−0.00	180
Robot x wages	−0.007550	−0.007550	0.00	173
Software x employment	−0.006183	−0.006183	0.00	269
Software x wages	−0.005049	−0.005049	0.00	268

Notes: The table verifies that the occupation-level contributions sum to the estimated automation exposure by minimum-wage interaction coefficient. Differences are numerical rounding error.

Table 10: Occupations with Highest Automation Exposure

Technology	Occupation	Exposure	States	Workers
<i>Software</i>				
Software	Packers and packagers by hand	100	48	293,714
Software	Dispatchers	100	42	89,627
Software	Water and sewage treatment plant operators	100	42	50,425
Software	Pest control occupations	100	29	49,144
Software	Power plant operators	100	36	26,935
Software	Locomotive operators: engineers and firemen	100	19	18,770
Software	Parking lot attendants	100	10	2,997
Software	Radiologic technologists and technicians	99	45	189,468
Software	Misc. material moving equipment operators	99	34	50,808
Software	Elevator installers and repairers	99	17	13,235
<i>Robots</i>				
Robots	Janitors	98	46	154,028
Robots	Elevator installers and repairers	98	1	352
Robots	Packers and packagers by hand	97	45	175,324
Robots	Dental laboratory and medical appliance technicians	97	12	14,183
Robots	Housekeepers, maids, butlers, and cleaners	97	6	1,626
Robots	Painting and decoration occupations	96	32	61,093
Robots	Machine feeders and offbearers	96	20	19,024
Robots	Structural metal workers	96	23	9,984
Robots	Locomotive operators: engineers and firemen	96	2	593
Robots	Excavating and loading machine operators	96	2	491

Notes: The table ranks occupations by automation exposure. This is an exposure ranking only; it does not identify which occupations drive the estimated interaction coefficient.

Wage results. Table 12 focuses only on the wage outcome. The wage interactions are central to the interpretation of both software and robot exposure.

High-wage downward contributors. Table 13 reports high-wage occupations that also push the interaction downward. It shows why the low- and middle-wage filter is important for interpreting the minimum-wage mechanism.

All negative contributors. Table 14 reports the largest negative contributors without applying the wage-band filter. This is a broader diagnostic table.

Table 11: Low- and Middle-Wage Occupations Pushing the Interaction Downward

Specification	Occupation	Signed contribution	Share of negative coefficient (%)	Exposure	Mean wage	Wage band
<i>Software x employment</i>						
Software x employment	Farm workers, incl. nursery farming, and marine life cultivation workers	-0.0026	41.63%	21	490	Low
Software x employment	Assemblers of electrical equipment	-0.0012	20.17%	89	673	Middle
Software x employment	Cashiers	-0.0010	16.73%	14	478	Low
Software x employment	Food preparation workers	-0.0007	12.09%	5	422	Low
Software x employment	Sheriffs, bailiffs, correctional institution officers	-0.0007	10.66%	81	789	Middle
Software x employment	Janitors	-0.0006	10.41%	83	572	Middle
Software x employment	Insurance adjusters, examiners, and investigators	-0.0005	7.69%	70	750	Middle
Software x employment	Radiologic technologists and technicians	-0.0004	7.28%	99	713	Middle
<i>Software x wages</i>						
Software x wages	Machine operators, n.e.c.	-0.0006	12.62%	93	723	Middle
Software x wages	Janitors	-0.0005	9.34%	83	570	Middle
Software x wages	Laborers, freight, stock, and material handlers, n.e.c.	-0.0004	8.15%	87	689	Middle
Software x wages	Social workers	-0.0004	7.47%	14	750	Middle
Software x wages	Packers and packagers by hand	-0.0003	6.61%	100	578	Middle
Software x wages	Secretaries and administrative assistants	-0.0003	5.66%	35	552	Low
Software x wages	Managers of properties and real estate	-0.0002	4.82%	16	815	Middle
Software x wages	Writers and authors	-0.0002	4.76%	1	702	Middle
<i>Robot x employment</i>						
Robot x employment	Laborers, freight, stock, and material handlers, n.e.c.	-0.0029	119.04%	93	699	Middle
Robot x employment	Assemblers of electrical equipment	-0.0026	104.76%	77	674	Middle
Robot x employment	Customer service reps, invest., adjusters, excl. insur.	-0.0013	53.58%	17	748	Middle
Robot x employment	Door-to-door sales, street sales, and news vendors	-0.0010	40.51%	14	696	Middle
Robot x employment	Tailors, dressmakers, and sewers	-0.0008	34.51%	27	549	Low
Robot x employment	Machine feeders and offbearers	-0.0006	25.91%	96	644	Middle
Robot x employment	Janitors	-0.0004	17.05%	98	761	Middle
Robot x employment	Stock and inventory clerks	-0.0004	16.15%	10	754	Middle
<i>Robot x wages</i>						
Robot x wages	Machine operators, n.e.c.	-0.0014	18.79%	82	725	Middle
Robot x wages	Packers and packagers by hand	-0.0007	8.95%	97	575	Middle
Robot x wages	Bookkeepers and accounting and auditing clerks	-0.0005	5.98%	35	638	Middle
Robot x wages	Advertising and related sales jobs	-0.0004	5.36%	7	832	Middle
Robot x wages	Janitors	-0.0004	5.33%	98	764	Middle
Robot x wages	Secretaries and administrative assistants	-0.0004	5.31%	44	615	Middle
Robot x wages	Laborers, freight, stock, and material handlers, n.e.c.	-0.0004	4.94%	93	700	Middle
Robot x wages	Painting and decoration occupations	-0.0003	4.31%	96	758	Middle

Notes: The table reports occupations with negative signed contributions. Mean wage is the weighted mean 1980 weekly wage. Low and middle wage bands are based on the exact weighted cutoffs used in the diagnostic: low if wage is at or below \$566.67, middle if wage is above \$566.67 and at or below \$856.79, and high if wage is above \$856.79.

Table 12: Low- and Middle-Wage Occupations Pushing Wage Interactions Downward

Specification	Occupation	Signed contribution	Share of negative coefficient (%)	Exposure	Mean wage	Wage band
<i>Software x wages</i>						
Software x wages	Machine operators, n.e.c.	-0.0006	12.62%	93	723	Middle
Software x wages	Janitors	-0.0005	9.34%	83	570	Middle
Software x wages	Laborers, freight, stock, and material handlers, n.e.c.	-0.0004	8.15%	87	689	Middle
Software x wages	Social workers	-0.0004	7.47%	14	750	Middle
Software x wages	Packers and packagers by hand	-0.0003	6.61%	100	578	Middle
Software x wages	Secretaries and administrative assistants	-0.0003	5.66%	35	552	Low
Software x wages	Managers of properties and real estate	-0.0002	4.82%	16	815	Middle
Software x wages	Writers and authors	-0.0002	4.76%	1	702	Middle
Software x wages	Welders, solderers, and metal cutters	-0.0002	4.19%	73	839	Middle
Software x wages	Shipping and receiving clerks	-0.0002	3.73%	91	681	Middle
<i>Robot x wages</i>						
Robot x wages	Machine operators, n.e.c.	-0.0014	18.79%	82	725	Middle
Robot x wages	Packers and packagers by hand	-0.0007	8.95%	97	575	Middle
Robot x wages	Bookkeepers and accounting and auditing clerks	-0.0005	5.98%	35	638	Middle
Robot x wages	Advertising and related sales jobs	-0.0004	5.36%	7	832	Middle
Robot x wages	Janitors	-0.0004	5.33%	98	764	Middle
Robot x wages	Secretaries and administrative assistants	-0.0004	5.31%	44	615	Middle
Robot x wages	Laborers, freight, stock, and material handlers, n.e.c.	-0.0004	4.94%	93	700	Middle
Robot x wages	Painting and decoration occupations	-0.0003	4.31%	96	758	Middle
Robot x wages	Welders, solderers, and metal cutters	-0.0003	3.59%	70	811	Middle
Robot x wages	Textile sewing machine operators	-0.0003	3.44%	74	414	Low

Notes: The table reports occupations with negative signed contributions. Mean wage is the weighted mean 1980 weekly wage. Low and middle wage bands are based on the exact weighted cutoffs used in the diagnostic: low if wage is at or below \$566.67, middle if wage is above \$566.67 and at or below \$856.79, and high if wage is above \$856.79.

Table 13: High-Wage Occupations Excluded from the Low- and Middle-Wage Diagnostic

Specification	Occupation	Signed contribution	Share of negative coefficient (%)	Exposure	Mean wage	Wage band
<i>Software x employment</i>						
Software x employment	Real estate sales occupations	-0.0013	20.57%	85	1,011	High
Software x employment	Sales occupations and sales representatives	-0.0006	9.26%	2	960	High
Software x employment	Accountants and auditors	-0.0005	8.60%	8	914	High
Software x employment	Managers of medicine and health occupations	-0.0004	6.96%	78	1,005	High
Software x employment	Police and detectives, public service	-0.0003	4.49%	60	956	High
Software x employment	Vocational and educational counselors	-0.0003	4.47%	11	876	High
Software x employment	Machinists	-0.0003	4.36%	93	881	High
Software x employment	Mail carriers for postal service	-0.0002	3.77%	1	990	High
<i>Software x wages</i>						
Software x wages	Driver/sales workers and truck Drivers	-0.0008	15.33%	66	915	High
Software x wages	Primary school teachers	-0.0004	8.28%	23	871	High
Software x wages	Accountants and auditors	-0.0004	7.83%	8	914	High
Software x wages	Fire inspection, fire fighting, and fire prevention occupations	-0.0002	3.65%	2	972	High
Software x wages	Secondary school teachers	-0.0002	3.42%	27	919	High
Software x wages	Managers of medicine and health occupations	-0.0002	3.18%	78	1,005	High
Software x wages	Subject instructors, college	-0.0001	2.28%	1	1,163	High
Software x wages	Heavy equipment and farm equipment mechanics	-0.0001	2.17%	97	892	High
<i>Robot x employment</i>						
Robot x employment	Sales occupations and sales representatives	-0.0037	150.05%	2	991	High
Robot x employment	Accountants and auditors	-0.0011	46.03%	8	919	High
Robot x employment	Editors and reporters	-0.0011	43.57%	18	941	High
Robot x employment	Engineering and science technicians	-0.0009	36.77%	89	890	High
Robot x employment	Mechanics and repairers, n.e.c.	-0.0006	24.08%	91	869	High
Robot x employment	Electrical engineers	-0.0004	17.79%	40	1,146	High
Robot x employment	Aircraft mechanics	-0.0003	12.11%	77	923	High
Robot x employment	Management analysts	-0.0002	9.64%	25	889	High
<i>Robot x wages</i>						
Robot x wages	Production supervisors or foremen	-0.0017	22.70%	38	963	High
Robot x wages	Sales occupations and sales representatives	-0.0015	20.38%	2	994	High
Robot x wages	Driver/sales workers and truck Drivers	-0.0009	11.97%	84	859	High
Robot x wages	Accountants and auditors	-0.0007	8.78%	8	920	High
Robot x wages	Office supervisors	-0.0005	6.23%	11	888	High
Robot x wages	Technical writers	-0.0002	2.33%	21	1,016	High
Robot x wages	Chemists	-0.0001	1.85%	21	970	High
Robot x wages	Sales supervisors and proprietors	-0.0001	1.82%	22	973	High

Notes: The table reports occupations with negative signed contributions. Mean wage is the weighted mean 1980 weekly wage. Low and middle wage bands are based on the exact weighted cutoffs used in the diagnostic: low if wage is at or below \$566.67, middle if wage is above \$566.67 and at or below \$856.79, and high if wage is above \$856.79.

Table 14: Occupations Pushing the Automation-Minimum Wage Interaction Downward

Specification	Occupation	Signed contribution	Share of negative coefficient (%)	Exposure	States
<i>Software x employment</i>					
Software x employment	Farm workers, incl. nursery farming, and marine life cultivation workers	-0.0026	41.63%	21	50
Software x employment	Real estate sales occupations	-0.0013	20.57%	85	47
Software x employment	Assemblers of electrical equipment	-0.0012	20.17%	89	46
Software x employment	Cashiers	-0.0010	16.73%	14	51
Software x employment	Food preparation workers	-0.0007	12.09%	5	39
Software x employment	Sheriffs, bailiffs, correctional institution officers	-0.0007	10.66%	81	49
Software x employment	Janitors	-0.0006	10.41%	83	51
Software x employment	Sales occupations and sales representatives	-0.0006	9.26%	2	51
<i>Software x wages</i>					
Software x wages	Driver/sales workers and truck Drivers	-0.0008	15.33%	66	51
Software x wages	Machine operators, n.e.c.	-0.0006	12.62%	93	48
Software x wages	Janitors	-0.0005	9.34%	83	51
Software x wages	Primary school teachers	-0.0004	8.28%	23	50
Software x wages	Laborers, freight, stock, and material handlers, n.e.c.	-0.0004	8.15%	87	51
Software x wages	Accountants and auditors	-0.0004	7.83%	8	51
Software x wages	Social workers	-0.0004	7.47%	14	50
Software x wages	Packers and packagers by hand	-0.0003	6.61%	100	45
<i>Robot x employment</i>					
Robot x employment	Sales occupations and sales representatives	-0.0037	150.05%	2	43
Robot x employment	Laborers, freight, stock, and material handlers, n.e.c.	-0.0029	119.04%	93	48
Robot x employment	Assemblers of electrical equipment	-0.0026	104.76%	77	46
Robot x employment	Customer service reps, invest., adjusters, excl. insur.	-0.0013	53.58%	17	32
Robot x employment	Accountants and auditors	-0.0011	46.03%	8	41
Robot x employment	Editors and reporters	-0.0011	43.57%	18	46
Robot x employment	Door-to-door sales, street sales, and news vendors	-0.0010	40.51%	14	29
Robot x employment	Engineering and science technicians	-0.0009	36.77%	89	40
<i>Robot x wages</i>					
Robot x wages	Production supervisors or foremen	-0.0017	22.70%	38	47
Robot x wages	Sales occupations and sales representatives	-0.0015	20.38%	2	42
Robot x wages	Machine operators, n.e.c.	-0.0014	18.79%	82	48
Robot x wages	Driver/sales workers and truck Drivers	-0.0009	11.97%	84	46
Robot x wages	Packers and packagers by hand	-0.0007	8.95%	97	41
Robot x wages	Accountants and auditors	-0.0007	8.78%	8	40
Robot x wages	Office supervisors	-0.0005	6.23%	11	35
Robot x wages	Bookkeepers and accounting and auditing clerks	-0.0005	5.98%	35	39

Notes: The table reports occupations with negative signed contributions, meaning occupations that push the interaction estimate downward. The share of negative coefficient can exceed 100 percent because positive occupation-level contributions offset negative contributions.

Absolute contributions. Table 15 ranks occupations by absolute contribution. This table includes both negative contributors and positive offsetting contributors, so it is useful for understanding which occupations have the most influence on the coefficient regardless of direction.

Table 15: Largest Absolute Occupation-Level Contributions to the Interaction Estimate

Specification	Occupation	Signed contribution	Share of absolute contribution (%)	Exposure	States
<i>Software x employment</i>					
Software x employment	Laborers, freight, stock, and material handlers, n.e.c.	-0.0004	4.50%	87	51
Software x employment	Machine operators, n.e.c.	-0.0000	3.96%	93	49
Software x employment	Sales occupations and sales representatives	-0.0006	3.42%	2	51
Software x employment	Accountants and auditors	-0.0005	2.92%	8	51
Software x employment	Sales supervisors and proprietors	0.0002	2.86%	30	51
Software x employment	Cashiers	-0.0010	2.56%	14	51
Software x employment	Janitors	-0.0006	2.54%	83	51
Software x employment	Office supervisors	0.0005	2.48%	13	50
<i>Software x wages</i>					
Software x wages	Laborers, freight, stock, and material handlers, n.e.c.	-0.0004	4.53%	87	51
Software x wages	Sales occupations and sales representatives	0.0000	4.41%	2	51
Software x wages	Machine operators, n.e.c.	-0.0006	3.97%	93	48
Software x wages	Janitors	-0.0005	3.69%	83	51
Software x wages	Accountants and auditors	-0.0004	3.65%	8	51
Software x wages	Cashiers	-0.0001	3.43%	14	51
Software x wages	Assemblers of electrical equipment	-0.0002	2.74%	89	46
Software x wages	Sales supervisors and proprietors	0.0002	2.73%	30	51
<i>Robot x employment</i>					
Robot x employment	Machine operators, n.e.c.	0.0003	10.39%	82	48
Robot x employment	Sales occupations and sales representatives	-0.0037	8.65%	2	43
Robot x employment	Laborers, freight, stock, and material handlers, n.e.c.	-0.0029	7.11%	93	48
Robot x employment	Production supervisors or foremen	0.0007	6.78%	38	49
Robot x employment	Assemblers of electrical equipment	-0.0026	5.01%	77	46
Robot x employment	Managers and administrators, n.e.c.	0.0018	4.63%	28	48
Robot x employment	Accountants and auditors	-0.0011	4.14%	8	41
Robot x employment	Packers and packagers by hand	0.0002	3.05%	97	45
<i>Robot x wages</i>					
Robot x wages	Sales occupations and sales representatives	-0.0015	9.31%	2	42
Robot x wages	Machine operators, n.e.c.	-0.0014	8.73%	82	48
Robot x wages	Production supervisors or foremen	-0.0017	7.69%	38	47
Robot x wages	Managers and administrators, n.e.c.	0.0009	6.55%	28	48
Robot x wages	Laborers, freight, stock, and material handlers, n.e.c.	-0.0004	5.78%	93	48
Robot x wages	Assemblers of electrical equipment	-0.0001	5.10%	77	46
Robot x wages	Accountants and auditors	-0.0007	3.96%	8	40
Robot x wages	Stock and inventory clerks	-0.0001	3.44%	10	36

Notes: This table reports the largest absolute occupation-level contributions. Negative signed contributions push the interaction estimate downward, while positive signed contributions offset it. This diagnostic identifies which occupations contribute most to the coefficient after controls and fixed effects; it is not the same as ranking occupations by automation exposure alone.

References

- Aaronson, Daniel, Eric French, Isaac Sorkin, and Ted To. 2018. “Industry Dynamics and the Minimum Wage: A Putty-Clay Approach.” *International Economic Review* 59(1): 51–84.
- Aaronson, Daniel, and Brian J. Phelan. 2018. “Wage Shocks and the Technological Substitution of Low-Wage Jobs.” *Economic Journal* 129(617): 1–34.
- Aaronson, Daniel, and Brian J. Phelan. 2019. “Wage shocks and the technological substitution of low-wage jobs.” *Economic Journal* 129(617): 1–34.
- Acemoglu, Daron, and David Autor. 2011. “Skills, Tasks and Technologies: Implications for Employment and Earnings.” In *Handbook of Labor Economics*, Vol. 4B, edited by Orley Ashenfelter and David Card, 1043–1171. Elsevier.
- Acemoglu, Daron, and Jörn-Steffen Pischke. 2003. “Minimum Wages and On-the-Job Training.” *Research in Labor Economics* 22: 159–202.
- Acemoglu, Daron, and Pascual Restrepo. 2019. “Automation and New Tasks: How Technology Displaces and Reinstates Labor.” *Journal of Economic Perspectives* 33(2): 3–30.
- Acemoglu, Daron, and Pascual Restrepo. 2020. “Robots and Jobs: Evidence from US Labor Markets.” *Journal of Political Economy* 128(6): 2188–2244.
- Acemoglu, Daron, and Pascual Restrepo. 2022. “Tasks, Automation, and the Rise in US Wage Inequality.” *Econometrica* 90(5): 1973–2016.
- Adams-Prassl, Abi, Teodora Boneva, Marta Golin, and Christopher Rauh. 2020. “Inequality in the Impact of the Coronavirus Shock: Evidence from Real Time Surveys.” *Journal of Public Economics* 189: 104245.
- Aghion, Philippe. 2002. “Schumpeterian Growth Theory and the Dynamics of Income Inequality.” *Econometrica* 70(3): 855–882.

- Austin, Benjamin, Edward Glaeser, and Lawrence Summers. 2018. “Jobs for the Heartland: Place-Based Policies in 21st-Century America.” *Brookings Papers on Economic Activity*: 151–255.
- Autor, David H. 2015. “Why Are There Still So Many Jobs? The History and Future of Workplace Automation.” *Journal of Economic Perspectives* 29(3): 3–30.
- Autor, David H. 2019. “Work of the Past, Work of the Future.” *AEA Papers and Proceedings* 109: 1–32.
- Autor, David H., and David Dorn. 2013. “The Growth of Low-Skill Service Jobs and the Polarization of the US Labor Market.” *American Economic Review* 103(5): 1553–1597.
- Autor, David H., David Dorn, and Gordon H. Hanson. 2013. “The China Syndrome: Local Labor Market Effects of Import Competition in the United States.” *American Economic Review* 103(6): 2121–2168.
- Autor, David H., David Dorn, and Gordon H. Hanson. 2015. “Untangling Trade and Technology: Evidence from Local Labour Markets.” *The Economic Journal* 125(584): 621–646.
- Autor, David H., Lawrence F. Katz, and Melissa S. Kearney. 2008. “Trends in U.S. Wage Inequality: Revising the Revisionists.” *Review of Economics and Statistics* 90(2): 300–323.
- Autor, David H., Frank Levy, and Richard J. Murnane. 2003. “The Skill Content of Recent Technological Change: An Empirical Exploration.” *Quarterly Journal of Economics* 118(4): 1279–1333.
- Autor, David H., Alan Manning, and Christopher L. Smith. 2016. “The Contribution of the Minimum Wage to US Wage Inequality over Three Decades: A Reassessment.” *American Economic Journal: Applied Economics* 8(1): 58–99.
- Azar, José, Ioana Marinescu, and Marshall we. Steinbaum. 2020. “Labor Market Concentration.” *Journal of Human Resources* 57(S): S167–S199.

- Brynjolfsson, Erik, Tom Mitchell, and Daniel Rock. 2018. “What Can Machines Learn, and What Does It Mean for Occupations and the Economy?” *AEA Papers and Proceedings* 108: 43–47.
- Card, David, and Alan B. Krueger. 1994. “Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania.” *American Economic Review* 84(4): 772–793.
- Cengiz, Doruk, Arindrajit Dube, Attila Lindner, and Ben Zipperer. 2019. “The Effect of Minimum Wages on Low-Wage Jobs.” *Quarterly Journal of Economics* 134(3): 1405–1454.
- Clemens, Jeffrey, and Michael Wither. 2021. “The Minimum Wage and the Great Recession: Evidence of Effects on the Employment and Income Trajectories of Low-Skilled Workers.” *Journal of Public Economics* 194: 104355.
- Dauth, Wolfgang, Sebastian Findeisen, Jens Suedekum, and Nicole Woessner. 2021. “The Adjustment of Labor Markets to Robots.” *Journal of the European Economic Association* 19(6): 3104–3153.
- Davis, Steven J., John C. Haltiwanger, and Scott Schuh. 1996. *Job Creation and Destruction*. Cambridge, MA: MIT Press.
- Deming, David J. 2017. “The Growing Importance of Social Skills in the Labor Market.” *Quarterly Journal of Economics* 132(4): 1593–1640.
- Dorn, David. 2009. “Essays on Inequality, Spatial Interaction, and the Demand for Skills.” Dissertation, University of St. Gallen.
- Dube, Arindrajit. 2019. “Impacts of Minimum Wages: Review of the International Evidence.” Working Paper. HM Treasury, UK.

- Dube, Arindrajit, T. William Lester, and Michael Reich. 2010. “Minimum Wage Effects Across State Borders: Estimates Using Contiguous Counties.” *Review of Economics and Statistics* 92(4): 945–964.
- Eckardt, Marcel Steffen. 2022. “Minimum wages in an automating economy.” *Journal of Public Economic Theory* 24(1): 58–91.
- Farber, Henry S., Daniel Herbst, Ilyana Kuziemko, and Suresh Naidu. 2021. “Unions and Inequality Over the Twentieth Century: New Evidence from Survey Data.” *Quarterly Journal of Economics* 136(3): 1325–1385.
- Felten, Edward W., Manav Raj, and Robert Seamans. 2018. “A Method to Link Advances in Artificial Intelligence to Occupational Abilities.” *AEA Papers and Proceedings* 108: 54–57.
- Firpo, Sergio, Nicole M. Fortin, and Thomas Lemieux. 2011. “Occupational Tasks and Changes in the Wage Structure.” Working Paper.
- Fortin, Nicole M., and Thomas Lemieux. 2012. “Changes in Wage Inequality in Canada: An Interprovincial Perspective.” *Canadian Journal of Economics* 45(2): 560–584.
- Harasztosi, Péter, and Attila Lindner. 2019. “Who Pays for the Minimum Wage?” *American Economic Review* 109(8): 2693–2727.
- Hyman, Benjamin G. 2022. “Can Displaced Labor Be Retrained? Evidence from Quasi-Random Assignment to Trade Adjustment Assistance.” Working Paper No. 22-05. Center for Economic Studies, U.S. Census Bureau.
- Katz, Lawrence F., and Alan B. Krueger. 2019. “The Rise and Nature of Alternative Work Arrangements in the United States, 1995–2015.” *Industrial and Labor Relations Review* 72(2): 382–416.
- Katz, Lawrence F., and Kevin M. Murphy. 1992. “Changes in Relative Wages, 1963–1987: Supply and Demand Factors.” *Quarterly Journal of Economics* 107(1): 35–78.

- Kletzer, Lori G., and Robert E. Litan. 2001. “A Prescription to Relieve Worker Anxiety.” Policy Brief No. 73. Brookings Institution.
- Lee, David S. 1999. “Wage Inequality in the United States During the 1980s: Rising Dispersion or Falling Minimum Wage?” *Quarterly Journal of Economics* 114(3): 977–1023.
- Lordan, Grace, and David Neumark. 2018. “People Versus Machines: The Impact of Minimum Wages on Automatable Jobs.” *Labour Economics* 52: 40–53.
- Manning, Alan. 2003. *Monopsony in Motion: Imperfect Competition in Labor Markets*. Princeton: Princeton University Press.
- Manning, Alan. 2021. “The Elusive Employment Effect of the Minimum Wage.” *Journal of Economic Perspectives* 35(1): 3–26.
- Mas, Alexandre, and Amanda Pallais. 2019. “Labor Supply and the Value of Non-Work Time: Experimental Estimates from the Field.” *American Economic Review: Insights* 1(1): 111–126.
- Meer, Jonathan, and Jeremy West. 2016. “Effects of the Minimum Wage on Employment Dynamics.” *Journal of Human Resources* 51(2): 500–522.
- Neumark, David, and William L. Wascher. 2008. *Minimum Wages*. Cambridge, MA: MIT Press.
- Neumark, David, J.M. Ian Salas, and William Wascher. 2014. “Revisiting the Minimum Wage-Employment Debate: Throwing Out the Baby with the Bathwater?” *Industrial and Labor Relations Review* 67(3): 608–648.
- Ruggles, Steven, Sarah Flood, Ronald Goeken, Josiah Grover, Erin Meyer, Jose Pacas, and Matthew Sobek. 2024. *IPUMS USA: Version 9.0 [dataset]*. Minneapolis, MN: IPUMS. <https://doi.org/10.18128/D010.V9.0>.

- Simon, Kosali Ilayperuma, and Robert Kaestner. 2004. “Do Minimum Wages Affect Non-wage Job Attributes? Evidence on Fringe Benefits.” *ILR Review* 57(1): 52–70.
- Stigler, George J. 1946. “The Economics of Minimum Wage Legislation.” *American Economic Review* 36(3): 358–365.
- Webb, Michael. 2020. “The Impact of Artificial Intelligence on the Labor Market.” Working Paper.
- Dustmann, Christian, Attila Lindner, Uta Schönberg, Matthias Umkehrer, and Philipp vom Berge. 2022. “Reallocation Effects of the Minimum Wage.” *Quarterly Journal of Economics* 137(1): 267–328.
- Acemoglu, Daron, and Pascual Restrepo. 2019. “Artificial Intelligence, Automation, and Work.” In *The Economics of Artificial Intelligence: An Agenda*, edited by Ajay Agrawal, Joshua Gans, and Avi Goldfarb, 197–236. Chicago: University of Chicago Press.
- Acemoglu, Daron, David Autor, Jonathon Hazell, and Pascual Restrepo. 2022. “Artificial Intelligence and Jobs: Evidence from Online Vacancies.” *Journal of Labor Economics* 40(S1): S293–S340.
- Acemoglu, Daron. 2024. “The Simple Macroeconomics of AI.” *Economic Policy* 40(121): 13–58.
- Autor, David H., David Dorn, and Gordon H. Hanson. 2013. “The Geography of Trade and Technology Shocks in the United States.” *American Economic Review* 103(3): 220–225.
- Davis, Steven J., John C. Haltiwanger, and Scott Schuh. 1996. *Job Creation and Destruction*. Cambridge, MA: MIT Press.
- Eloundou, Tyna, Sam Manning, Pamela Mishkin, and Daniel Rock. 2023. “GPTs are GPTs: An Early Look at the Labor Market Impact Potential of Large Language Models.” arXiv preprint arXiv:2303.10130.

Felten, Edward W., Manav Raj, and Robert Seamans. 2023. “How will Language Models like ChatGPT Affect Occupations and Industries?” arXiv preprint arXiv:2303.01157.